

Lab 3 – Architectural Justification

Problem Statement #14 – Corporate Mental Wellness & Counseling Platform

SRN: PES1UG24CS620 | Section : 5K

Architecture Selection

We chose the **Microservices Architecture** for the Corporate Mental Wellness & Counseling Platform.

Architectural Choice

Layered was rejected because a single deployable unit couples the therapist-facing code to the same data tier that holds employee PII, and it cannot scale the scheduler on its own. **Client-Server** was rejected because a central server is a single point of failure and a scalability bottleneck for globally distributed offices.

Microservices gives small, independently deployable components (Identity & Pseudonym, Appointment & Session, Assessment & Wellness) behind an API Gateway, each with its own datastore, as shown in the component diagram.

Two Reasons for the Choice

1. Identity isolation supports FR-001 / NFR-002. The platform must never expose corporate email or employee ID to therapists. As a separate Identity & Pseudonym Service with its own encrypted Identity Vault, PII is confined to one small trust zone. The Gateway swaps the employee's identity for a pseudo-ID, so the Appointment and Wellness services (which therapists touch) only ever handle the pseudo-ID.

2. Independent scheduling for global peak load (FR-003 / NFR-001). Booking and timezone conversion for employees and therapists across offices is the busiest, most correctness-critical workflow. Placing it in its own Appointment & Session Service lets it be deployed, tuned and scaled separately from the low-traffic Stress Assessment and Wellness functions (FR-002, FR-005), which can also evolve without redeploying the scheduler.

Security Advantage

Data minimisation and reduced blast radius. Only the Identity Service can read the Identity Vault; all other services communicate over mTLS, use the pseudo-ID, and own separate databases. A compromise of the therapist-facing Appointment service, or its database, therefore cannot reveal an employee's corporate identity. The API Gateway is the single entry point that enforces role-based access control (Employee vs Therapist), giving one place to audit and protect counseling data.

Performance Benefit

Targeted horizontal scaling and fault isolation. Stateless services behind the Gateway can be replicated on demand: during peak booking periods only the Appointment & Session Service is scaled out, avoiding the cost of scaling the whole application. Slot lists and timezone rules can be cached inside that service, and a failure or slowdown in the Wellness service does not stop employees from booking sessions.

Trade-offs Acknowledged

Microservices add network latency, operational complexity and eventual-consistency concerns (e.g. the Wellness service reads booking history through the read-only IBookingHistory interface). These are mitigated by the API Gateway, health checks and simple read-only inter-service contracts, and are outweighed by the confidentiality and scalability needs of this platform.